Introduction

This topic covers Activation Dynamics and Neural Network Models used in Artificial Neural Networks (ANN) to mathematically describe how neuron activations and synaptic weights evolve over time. These models are based on differential equations and are inspired by biological neurons. They explain important behaviors such as decay, excitation, inhibition, feedback, associative memory, shunting mechanisms, and synaptic learning. Such formulations are fundamental in understanding Hopfield networks, associative memories, shunting models, and competitive neural systems.

Equations

1) Activation Dynamics

$$\dot{x}_i = \frac{dx_i}{dt} = h(\cdot)$$

Terms:

- x_i– Activation of neuron i
- $\dot{x}_i, \frac{dx_i}{dt}$– Rate of change of activation
- $h(\cdot)$– Activation dynamics function

Expanded form

$$\frac{dx_i}{dt} = -ax_i + \sum_j w_{ij}\, s_j + I_i$$

Terms:

- a– Decay constant
- w_{ij}– Synaptic weight from neuron j to neuron i
- s_j– Output of neuron j
- I_i– External input

2) Passive Decay Model

$$\frac{dx_i(t)}{dt} = -A_i x_i(t)$$

Terms:

- A_i– Passive decay (leakage) constant

$$x_i(t) = x_i(0)e^{-A_i t}$$

- $x_i(0)$– Initial activation
- t– Time

3) Modified Passive Decay Model

$$\frac{dx_i(t)}{dt} = -\frac{A_i}{C_i} x_i(t)$$

Terms:

- C_i– Capacitance/scaling constant

$$x_i(t) = x_i(0)e^{-\frac{A_i}{C_i}t}$$

4) Nonzero Resting Potential

$$\frac{dx_i(t)}{dt} = -A_i x_i(t) + p_i$$

Terms:

- p_i– Resting potential input

$$x_i(t) = \left(x_i(0) - \frac{p_i}{A_i}\right) e^{-A_i t} + \frac{p_i}{A_i}$$

Steady state

$$x_i(\infty) = \frac{p_i}{A_i}$$

- $x_i(\infty)$– Steady state value

5) Zero Resting Potential with External Input

$$\frac{dx_i(t)}{dt} = -A_i x_i(t) + B_i I_i$$

Terms:

- B_i– Excitatory scaling parameter
- I_i– External input

$$x_i(t) = x_i(0)e^{-A_i t} + \frac{B_i I_i}{A_i}(1 - e^{-A_i t})$$

Steady state

$$x_i(\infty) = \frac{B_i I_i}{A_i}$$

6) Additive Autoassociative Model

$$\frac{dx_i(t)}{dt} = -A_i x_i(t) + B_i I_i + \sum_{j=1}^{n} w_{ij}\, \phi_j(x_j(t))$$

Terms:

- w_{ij}– Synaptic weight
- $\phi_j(x_j(t))$– Output/activation function
- n– Number of neurons

7) Inhibitory Feedbacks & External Input

$$\frac{dx_i(t)}{dt} = -A_i x_i(t) - B_i I_i - \sum_{j \neq i} w_{ij}\, \phi_j(x_j(t))$$

Terms:

- $\sum_{j \neq i}$ – Summation over all neurons except i
- Negative signs – Inhibitory effects

8) Perkel Model

$$\frac{dx_i(t)}{dt} = -\frac{1}{R_i} x_i(t) + \sum_{j=1}^{n} \frac{1}{R_{ij}} \phi_j(x_j(t))$$

Terms:

- R_i– Membrane resistance
- R_{ij}– Coupling resistance

9) Hopfield Model

$$f(z) = \frac{1}{1 + e^{-z}}$$

Terms:

- z– Net input
- $f(z)$– Sigmoid activation function

10) Heteroassociative Networks

Layer–1

$$\frac{dx_i(t)}{dt} = -A_i x_i(t) + \sum_{j=1}^{N} f_j\,(y_j(t)) v_{ij} + I_i$$

Terms:

- v_{ij}– Forward connection weight
- $f_j(y_j(t))$– Output of layer–2 neuron
- N– Number of layer–2 neurons

Layer–2

$$\frac{dy_j(t)}{dt} = -B_j y_j(t) + \sum_{i=1}^{M} g_i\,(x_i(t)) w_{ji} + J_j$$

Terms:

- w_{ji}– Feedback weight
- $g_i(x_i(t))$– Output of layer–1 neuron
- J_j– External input
- M– Number of layer–1 neurons

11) Shunting Activation Model

$$\frac{dx_i(t)}{dt} = -A_i x_i(t) + [B_i - x_i(t)] I_i$$

Terms:

- $B_i - x_i(t)$– Excitatory shunting term

Steady state

$$x_i(\infty) = \frac{B_i I_i}{A_i + I_i}$$

12) On-Center Off-Surround

$$\dot{x}_i(t) = -A_i x_i(t) + [B_i - x_i(t)]I_i - x_i(t)\sum_{j \neq i} I_j$$

Terms:

- $\sum_{j \neq i} I_j$– Lateral inhibitory input

13) Shunting Activation Model with Lower Limit

$$\frac{dx_i(t)}{dt} = -A_i x_i(t) + [B_i - x_i(t)]I_i - [E_i + x_i(t)]\sum_{j \neq i} I_j$$

Terms:

- E_i– Lower bound parameter

14) Shunting Activation Model with Feedback

$$\dot{x}_i(t) = -A_i x_i(t) + (B_i - x_i)[I_i + f_i(x_i)] - (E_i + x_i)[J_i + \sum_{j \neq i} f_j\,(x_j) w_{ij}]$$

Terms:

- $f_i(x_i)$, $f_j(x_j)$– Activation functions
- w_{ij}– Feedback weight

15) Excitatory Term

$$(B_i - C_i x_i)[I_i + f_i(x_i)]$$

- C_i– Excitatory gain parameter

16) Inhibitory Term

$$(E_i + D_i x_i)[J_i + \sum_{j \neq i} f_j\,(x_j) w_{ij}]$$

- D_i– Inhibitory gain parameter

17) Synaptic Dynamics

$$\frac{dw_{ij}(t)}{dt} = -w_{ij}(t) + f_i(x_i(t)) f_j(x_j(t))$$

Terms:

- $w_{ij}(t)$– Time-varying synaptic weight
- $\frac{dw_{ij}}{dt}$– Rate of change of weight
- $f_i(x_i) f_j(x_j)$– Hebbian learning term